

The empirical recurrence for OEIS A254548, recovered from its tail

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Abstract

OEIS A254548 records “Empirical recurrence of order 91” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 487$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 96$. Its last coefficient is $c_{91} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-91 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254548 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock diagonal sum minus antidiagonal minimum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

12260, 34780, 129677, 380406, 1191782, 3475823, 10616893, 30951404, ...

The entry states:

Empirical recurrence of order 91 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:48:00, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 487$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 487$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 91: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 91.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 974$ exact terms from $n = 5$ on, it returns order 91 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{91} c_i a(n-i),$$

c_1	6	c_{24}	−67638	c_{47}	−36962	c_{70}	−170
c_2	−8	c_{25}	20357	c_{48}	7514	c_{71}	157
c_3	−12	c_{26}	57248	c_{49}	17297	c_{72}	319
c_4	37	c_{27}	−66983	c_{50}	−24146	c_{73}	−561
c_5	−48	c_{28}	98834	c_{51}	41410	c_{74}	90
c_6	47	c_{29}	−113894	c_{52}	−45215	c_{75}	−384
c_7	62	c_{30}	93177	c_{53}	32291	c_{76}	43
c_8	−136	c_{31}	−99013	c_{54}	−13422	c_{77}	49
c_9	−186	c_{32}	62026	c_{55}	4490	c_{78}	23
c_{10}	274	c_{33}	38738	c_{56}	6608	c_{79}	80
c_{11}	1023	c_{34}	−143622	c_{57}	−4066	c_{80}	51
c_{12}	−2268	c_{35}	156837	c_{58}	−7587	c_{81}	15
c_{13}	1439	c_{36}	−168319	c_{59}	−1459	c_{82}	−21
c_{14}	−549	c_{37}	126242	c_{60}	−689	c_{83}	−21
c_{15}	−1211	c_{38}	−21907	c_{61}	961	c_{84}	−22
c_{16}	5668	c_{39}	42423	c_{62}	7787	c_{85}	2
c_{17}	−13358	c_{40}	−22895	c_{63}	−4489	c_{86}	−3
c_{18}	21239	c_{41}	−50377	c_{64}	4376	c_{87}	5
c_{19}	−12778	c_{42}	66896	c_{65}	−3411	c_{88}	4
c_{20}	−2101	c_{43}	−92294	c_{66}	−1413	c_{89}	1
c_{21}	26181	c_{44}	107650	c_{67}	339	c_{90}	0
c_{22}	−62930	c_{45}	−79558	c_{68}	−802	c_{91}	−1
c_{23}	68781	c_{46}	45953	c_{69}	1880		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 96$, and it is the recurrence OEIS A254548 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{91} - c_1x^{91-1} - \dots - c_{91}$. Its constant term is $-c_{91} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 91 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 91, so $\deg q = 91$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 91, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 91, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-91 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A254548.
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